Machine Learning Overview

Data Science

COMP5122M

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What will you learn?

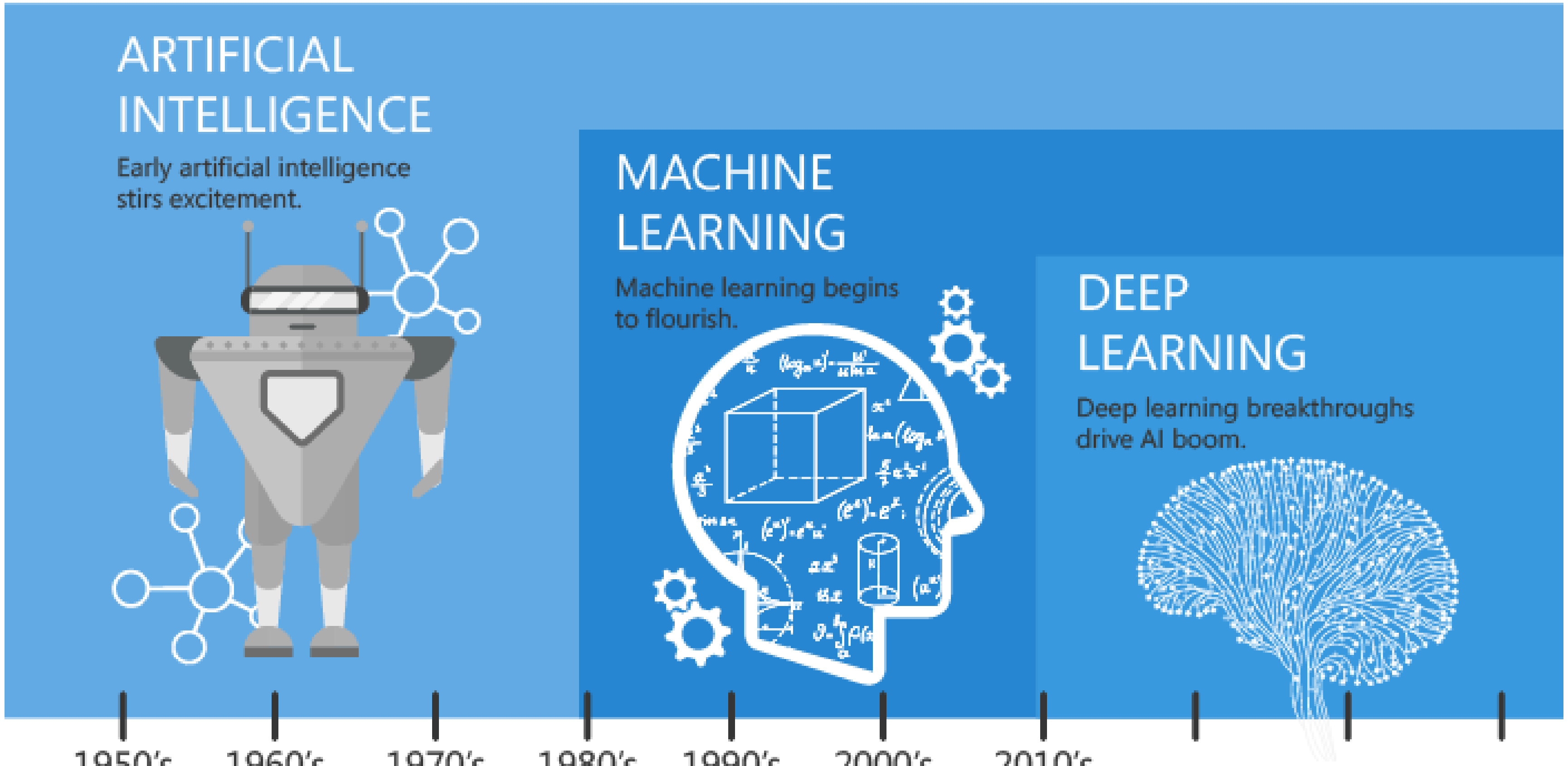
- What is Data Mining?
- What is Artificial Intelligence?
- What is Machine Learning
- Data Mining vs Machine Learning
- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

What is Data Mining?

- *Data mining, also known as knowledge discovery in data (KDD), is the process of uncovering patterns and other valuable information from large data sets.*

(source: IBM https://www.ibm.com/topics/data-mining

What is Artificial intelligence?



Since an early flush of optimism in the 1950's, smaller subsets of artificial intelligence - first machine learning, then deep learning, a subset of machine learning - have created ever larger disruptions.

What is Artificial intelligence?

- The term ***Artificial Intelligence***, as a research field, was coined at the conference on the campus of Dartmouth College in the summer of **1956**, even though the idea was around since antiquity.
- For instance, in the first manifesto of Artificial Intelligence, "*Intelligent Machinery*", in **1948** Alan Turing distinguished two different approaches to AI, which may be termed ***"top-down"*** or ***knowledge-driven AI*** and ***"bottom-up"*** or ***data-driven AI***

(sources: Wikipedia & http://www.alanturing.net & Stanford Encyclopedia of Philosophy)

What is Artificial intelligence?

- The two different approaches to AI can be detailed:
  - ***"top-down"*** or ***knowledge-driven AI***
    - cognition = high-level phenomenon, independent of low-level details of implementation mechanism, first neuron (1943), first neural network machine (1950), neucognitron (1975)
    - Evolutionary Algorithms (1954,1957, 1960), Reasoning (1959,1970), Expert Systems (1970), Logic, Intelligent Agent Systems (1990)…
  - ***"bottom-up" or data-driven AI***
    - opposite approach, start from data to build incrementally and mathematically mechanisms taking decisions
    - Machine learning algorithms, Decision Trees (1983), Backpropagation (1984-1986), Random Forest (1995), Support Vector Machine (1995), Boosting (1995), Deep Learning (1998/2006)…

(sources: Wikipedia & http://www.alanturing.net & Stanford Encyclopedia of Philosophy)

What is Artificial intelligence?

- *AI is originally defined, by Marvin Lee Minsky, as "the construction of computer programs doing tasks, that are, for the moment, accomplished more satisfyingly by human beings because they require high level mental processes such as:* ***learning****. perceptual organization of memory and critical reasoning".*

- There are so the "artificial" side with the usage of computers or sophisticated electronic processes and the side "intelligence" associated with its goal to imitate the (human) behavior.

(sources: Wikipedia & http://www.alanturing.net & Stanford Encyclopedia of Philosophy)

Why Learn?

- *Machine learning is programming computers to optimize a performance criterion using example data or past experience.*
- *There is no need to “learn” to calculate payroll*
- *Learning is used when:*
    - *Human expertise does not exist (navigating on Mars),*
    - *Humans are unable to explain their expertise (speech recognition)*
    - *Solution changes in time (routing on a computer network)*
    - *Solution needs to be adapted to particular cases (user biometrics)*

(sources: Ethem Alpaydın)

What do we mean by Learning?

- *Learning general models from a data of particular examples*
- *Data is cheap and abundant (data warehouses, data marts); knowledge is expensive and scarce.*
- *Example in retail: Customer transactions to consumer behavior:*
    - *People who bought "Da Vinci Code" also bought "The Five People You Meet in Heaven" (www.amazon.com)*
- *Build a model that is a good and useful approximation to the data.*

(sources: Ethem Alpaydın)

What do we mean by Learning?

- *Retail: Market basket analysis, Customer relationship management (CRM)*
- *Finance: Credit scoring, fraud detection*
- *Manufacturing: Optimization, troubleshooting*
- *Medicine: Medical diagnosis*
- *Telecommunications: Quality of service optimization*
- *Bioinformatics: Motifs, alignment*
- *Web mining: Search engines*

(sources: Ethem Alpaydın)

What is Machine Learning?

- *Optimize a performance criterion using example data or past experience.*
- *Role of Statistics: Inference from a sample*
- *Role of Computer science: Efficient algorithms to*
    - *Solve the optimization problem*
    - *Representing and evaluating the model for inference*

(sources: Ethem Alpaydın)

What is Machine Learning?

$$\left(x \right) \xrightarrow{f(\mathbf{X},a)\ ?} y$$

$\left(x \right)$

Face Detection

Betting on sports

Speech Recognition

y

Scores, ranking...

(sources: Frederic Precioso)

Problems

- Regression
- (Supervised) Classification
- Density Estimation / Clustering (Unsupervised Classification)
- Feature Selection
- Association analysis
- Anomality/Novelty/Drift
- …

Possible Solutions

- Machine Learning
  - Support Vector Machine
  - Artificial Neural Network
  - Boosting
  - Decision Tree
  - Random Forest
  - …
- Statistical Learning
  - Gaussian Models (GMM)
  - Naïve Bayes
  - Gaussian processes
  - …
- Other techniques

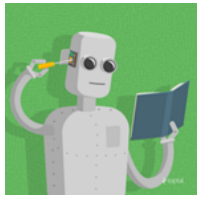
(sources: Frederic Precioso)

Data Mining vs Machine Learning

- *Data mining looks for patterns that already exist in the data, while machine learning learns from past data to predict future outcomes.*

	Data Mining	Machine Learning
Focus	Discovery of hidden patterns or knowledge from data	Development of algorithms that learn from data
Goal	Extract insights and information from existing datasets	Build models to make predictions or perform tasks
Usage	Identifying patterns, trends, and anomalies	Predictive modeling, classification, clustering, etc.

(sources: https://www.simplilearn.com/data-mining-vs-machine-learning-article)



Labeled Data

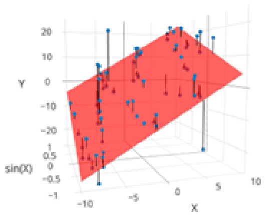
Unlabeled Data

Supervised Learning

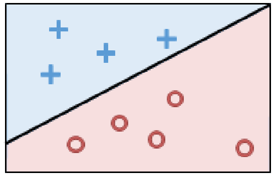
Quantitative Response

Categorical Response

Regression



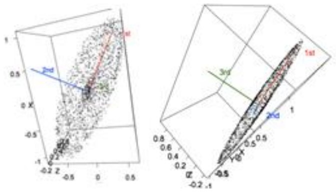
Classification



Nearest Neighbors
Logistic Regression

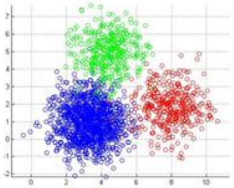
Unsupervised Learning

Dimensionality Reduction

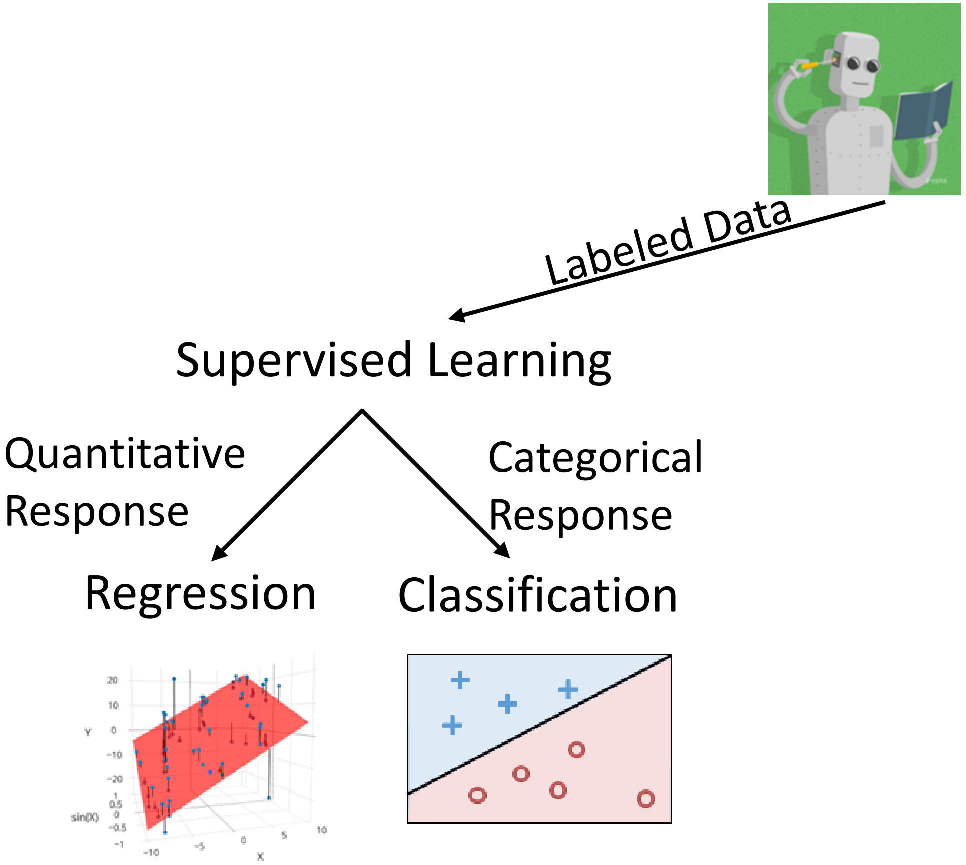


Principal Component Analysis

Clustering



KMeans Clustering
Agglomerative Clustering



In “Supervised Learning”:

- Goal is to create a function that maps inputs to outputs.
- Model is learned from example input/output pairs. Each pair consists of:
  - Input vector.
  - Output value (**label**).
- Regression: Output value is quantitative.
- Classification: Output value is categorical.

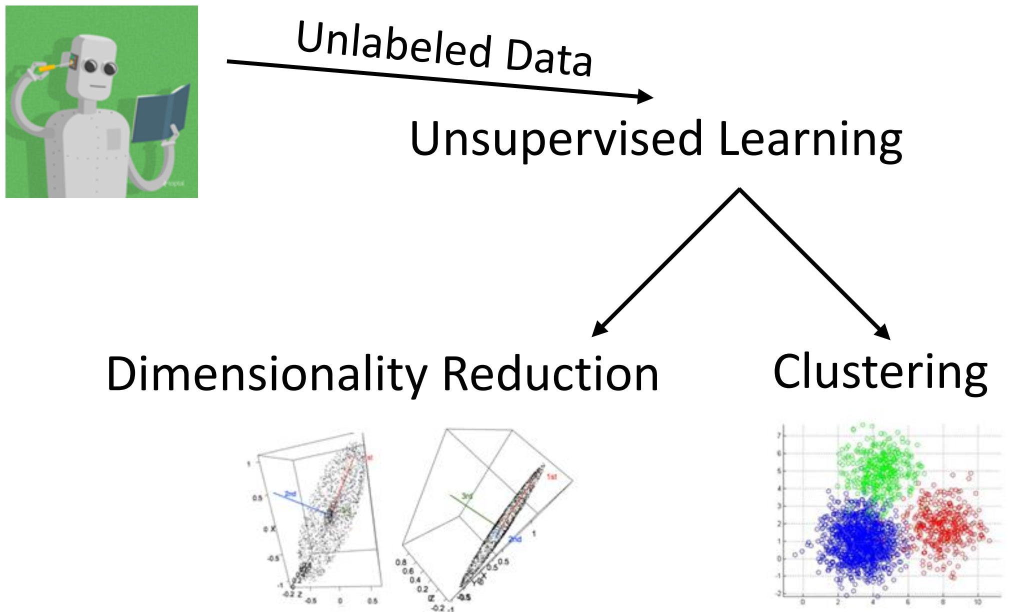
Supervised Learning

- Supervised machine learning is one of the most commonly used and successful types of machine learning.

- Supervised learning is used whenever we want to predict a certain outcome from a given input
    - We have examples of input/output pairs
    - We build a machine learning model from these input/output pairs -> training set
    - Our goal is to make accurate predictions for new, never-before-seen data

In “Unsupervised Learning”:

- Goal is to identify patterns in **unlabeled** data.
    - We do not have input/output pairs.
    - Note: Sometimes we may have labels, but we’re just choosing to ignore them (e.g. PCA on labeled data).



Dimensionality Reduction.

- Solution: PCA.

Unsupervised Learning

- Unsupervised learning includes all kinds of machine learning where there is no known output, no teacher to instruct the learning algorithm. -> no supervision

- In unsupervised learning, the learning algorithm is just shown the input data and asked to extract knowledge from this data.

Supervised Learning
Classification and Regression

- There are two major types of supervised machine learning problems, called classification and regression.

- In classification, the goal is to predict a class label, which is a choice from a predefined list of possibilities.
  - Binary classification
    - Classifying emails as either spam or not spam.
    - “Is this email spam?”: yes/no, positive/negative
  - Multiclass classification
    - Predicting what language a website is in: pre-defined list of possible languages

Unsupervised Learning
Transformations of the dataset

- **Unsupervised transformations** of a dataset are algorithms that **create a new representation of the data** which might be easier for humans or other machine learning algorithms to understand compared to the original representation of the data.

  For example,

  Dimensionality Reduction takes a high-dimensional representation of the data, consisting of many features, and finds a new way to represent this data that summarizes the essential characteristics with fewer features.

  - Dimensionality reduction to two dimensions for visualization purposes
  - Topic extraction on collections of text documents

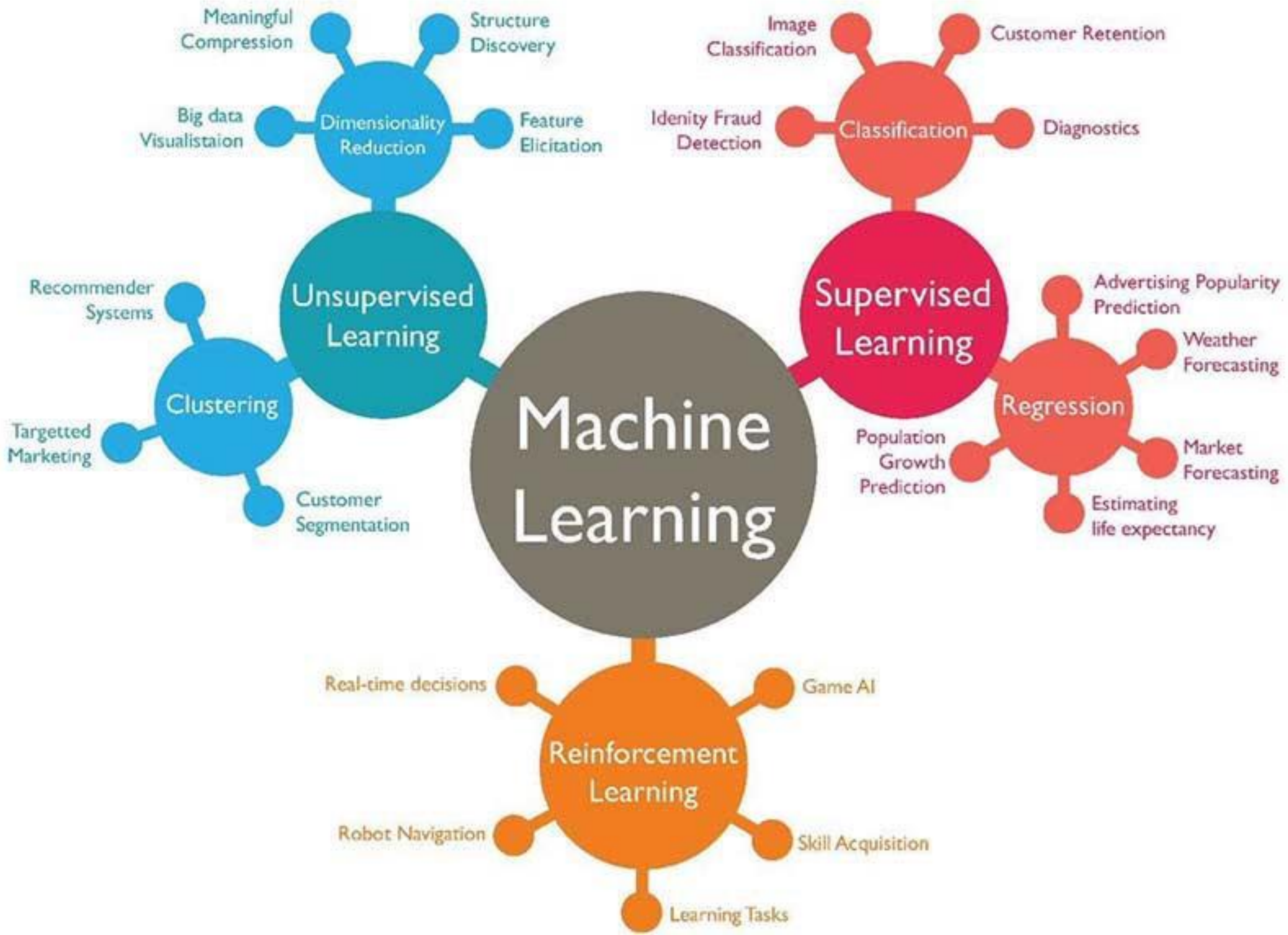
Unsupervised Learning
Clustering

- Clustering algorithms, on the other hand, partition data into distinct groups of similar items.

- The goal is to split up the data in such a way that points within a single cluster are very similar and points in different clusters are different.

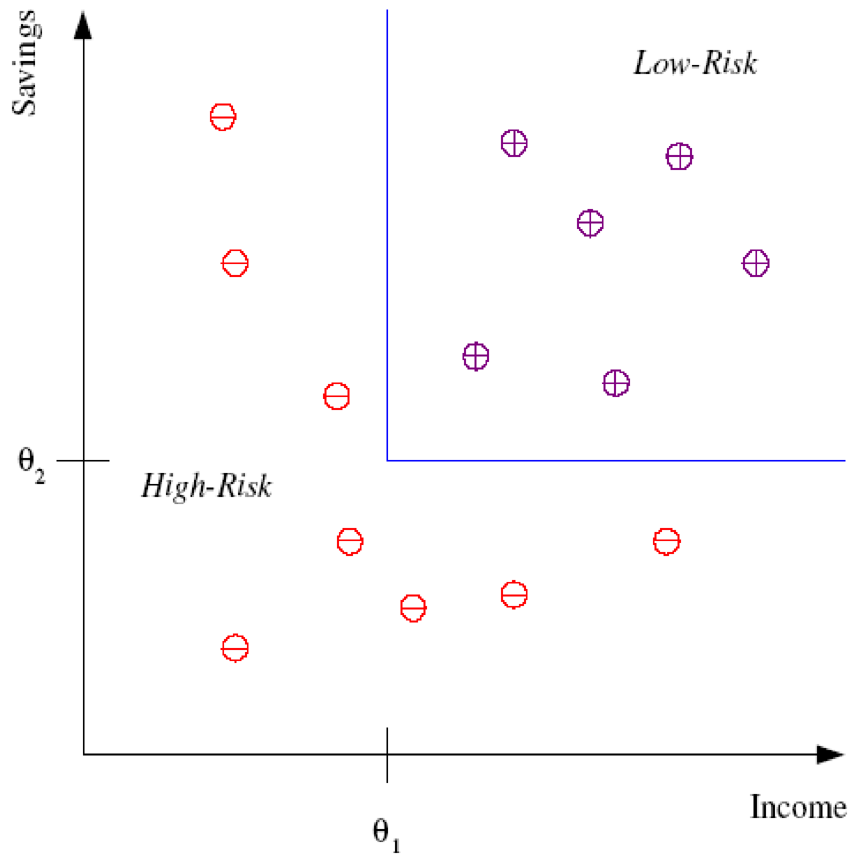
- Similarly to classification algorithms, clustering algorithms assign (or predict) a cluster to a particular data point.

For example: Organizing uploaded pictures, group together pictures that show the same person or location.


Machine Learning
Unsupervised Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Supervised Learning
Classification
Image Classification
Customer Retention
Idenity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Population Growth Prediction
Market Forecasting
Estimating life expectancy
Reinforcement Learning
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

Classification

- *Example: Credit scoring*
- *Differentiating between low-risk and high-risk customers from their income and savings*



Discriminant: IF income > θ1 AND savings > θ2
THEN low-risk ELSE high-risk

(sources: Ethem Alpaydın)

Classification: Applications

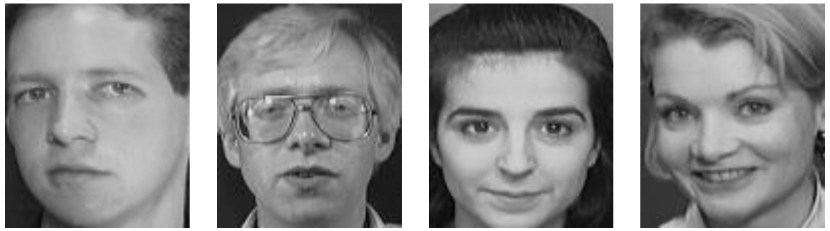
- *Also known as Pattern recognition*
- *Face recognition: Pose, lighting, occlusion (glasses, beard), make-up, hair style*
- *Character recognition: Different handwriting styles.*
- *Speech recognition: Temporal dependency.*
- *Use of a dictionary or the syntax of the language.*
- *Sensor fusion: Combine multiple modalities; eg, visual (lip image) and acoustic for speech*
- *Medical diagnosis: From symptoms to illnesses*

(sources: Ethem Alpaydın)

Classification: Face Recognition

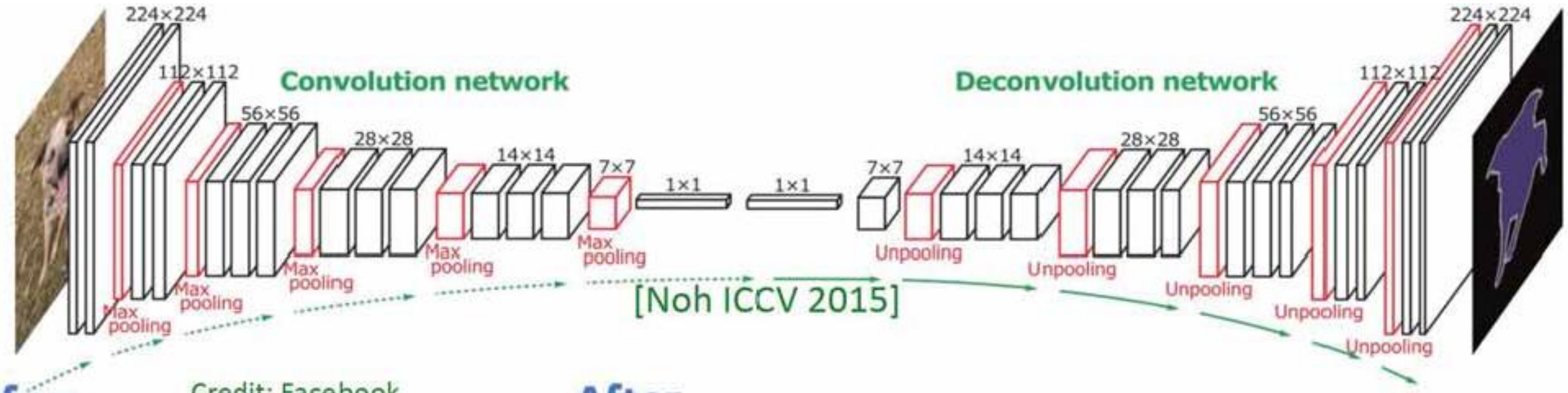
- *Training examples of a person*

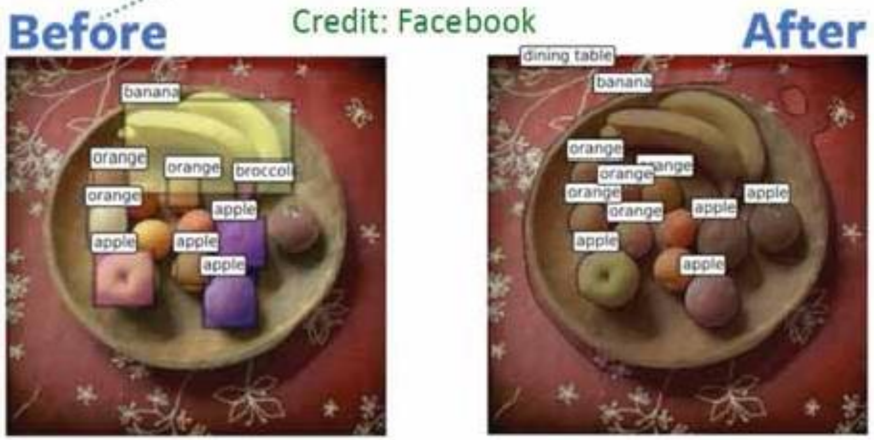
- *Test Images*



(sources: Ethem Alpaydın)

Supervised Image Segmentation

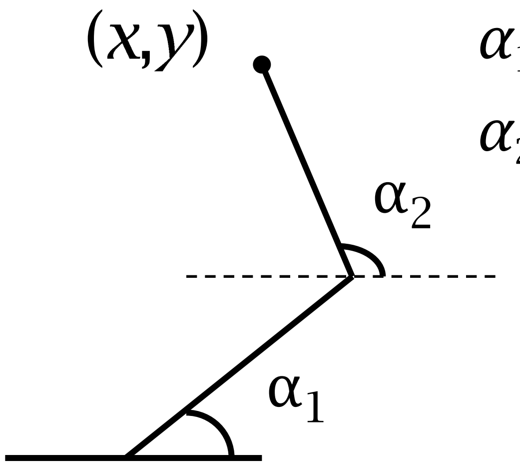




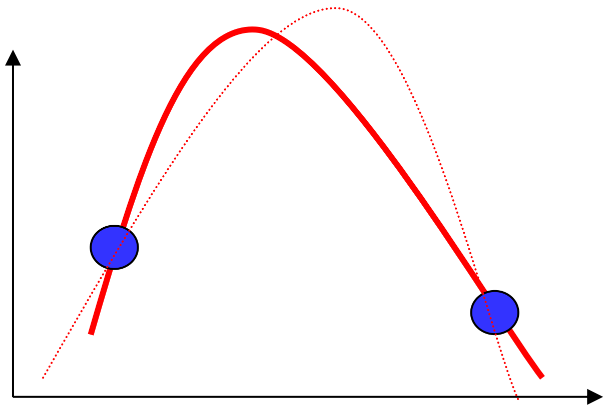
MS COCO Detection Challenge!

Regression: Applications

- *Navigating a car: Angle of the steering wheel (CMU NavLab)*
- *Kinematics of a robot arm*



$$\alpha_1 = g_1(x,y)$$
$$\alpha_2 = g_2(x,y)$$



Response surface design

(sources: Ethem Alpaydın)

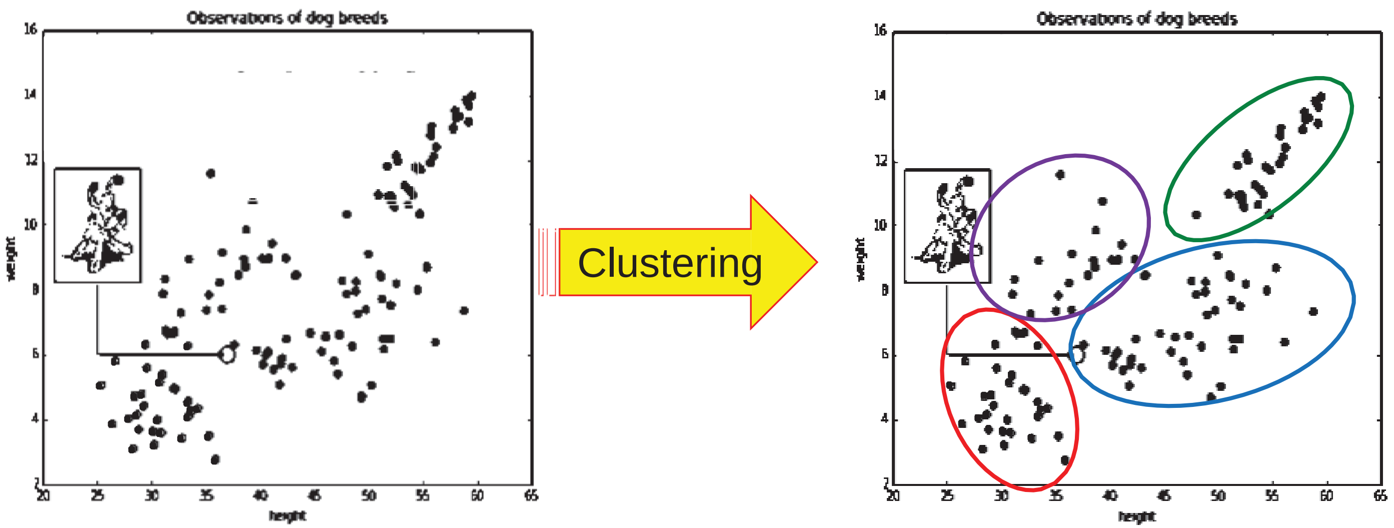
Unsupervised Learning

- *Learning "what normally happens"*
- *No output*
- *Clustering: Grouping similar instances*
- *Example applications*
  - *Customer segmentation in CRM*
  - *Image compression: Color quantization*
  - *Bioinformatics: Learning motifs*

(sources: Ethem Alpaydın)

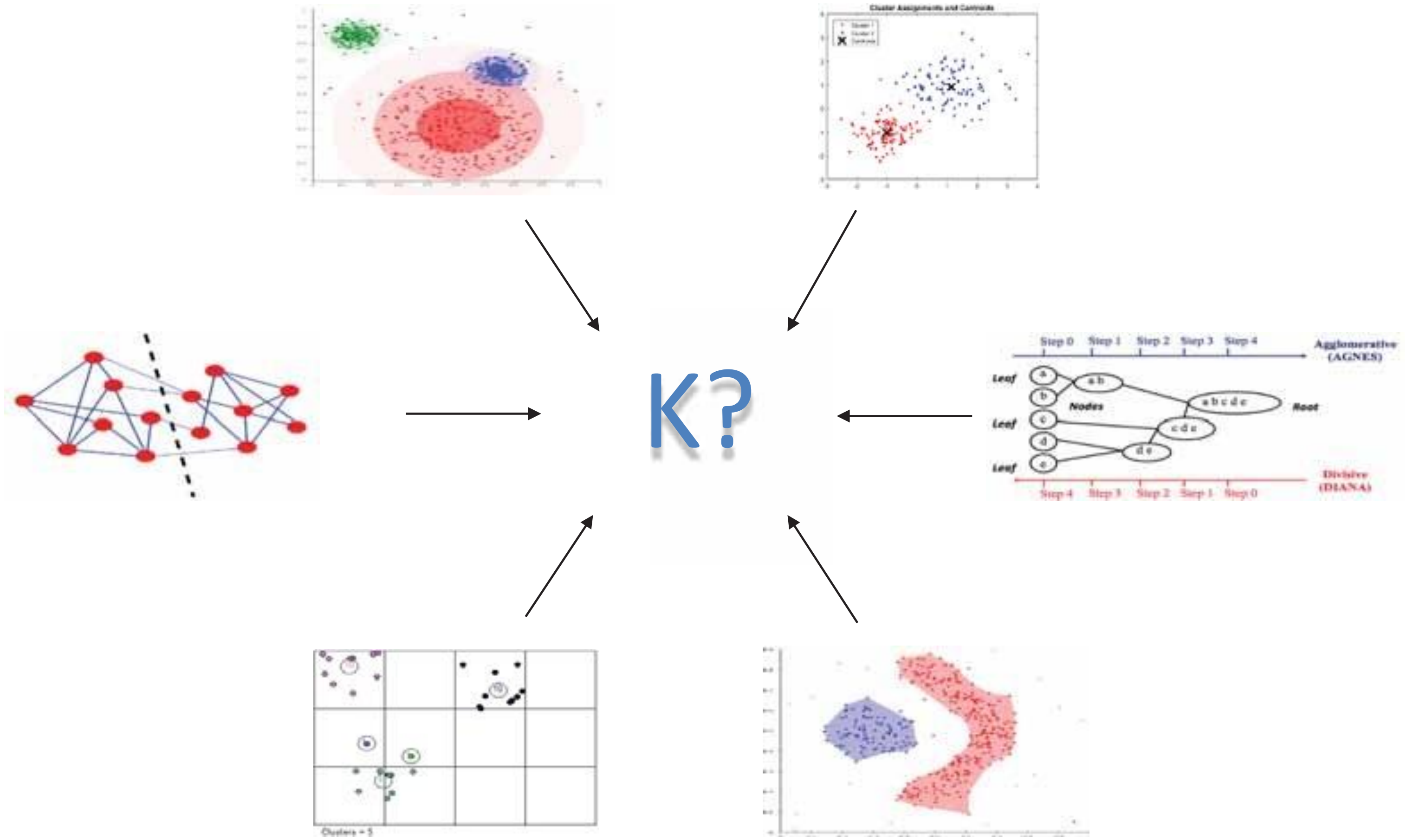
Clustering

Clustering: Partition a dataset into groups based on the similarity between the instances.



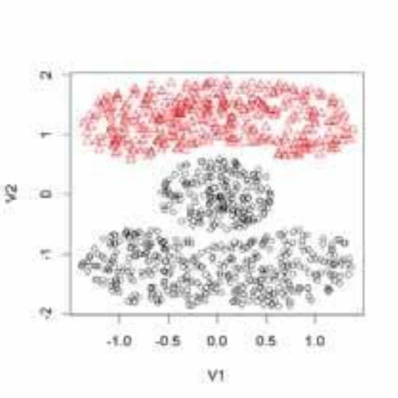
(sources: Frederic Precioso)

How many Clusters?

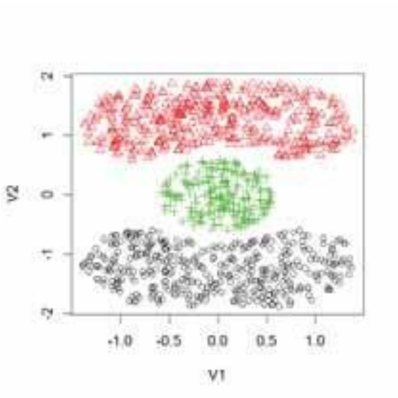


(sources: Frederic Precioso)

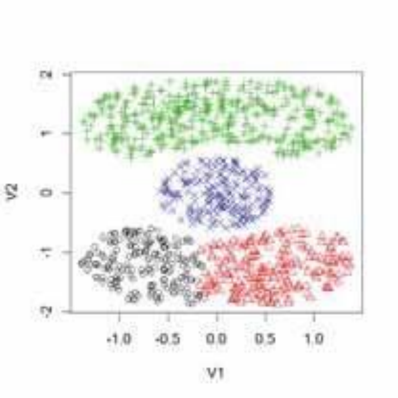
Algorithms’ Parameters



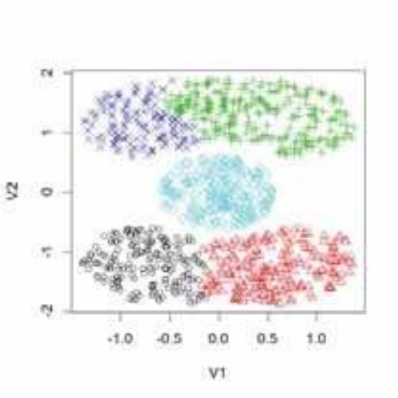
Avg. linkage: K=2



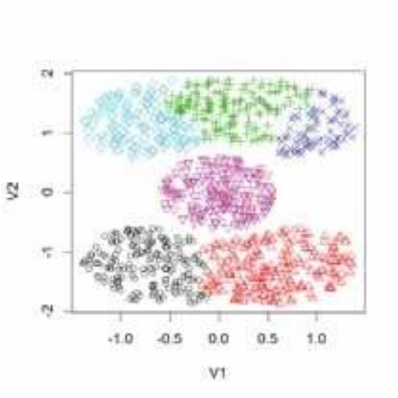
Avg. linkage: K=3



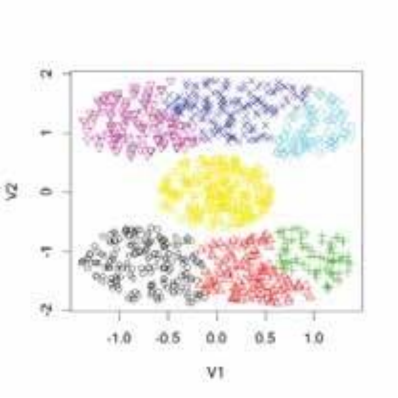
Avg. linkage: K=4



Avg. linkage: K=5



Avg. linkage: K=6



Avg. linkage: K=7

Which metric/distance?

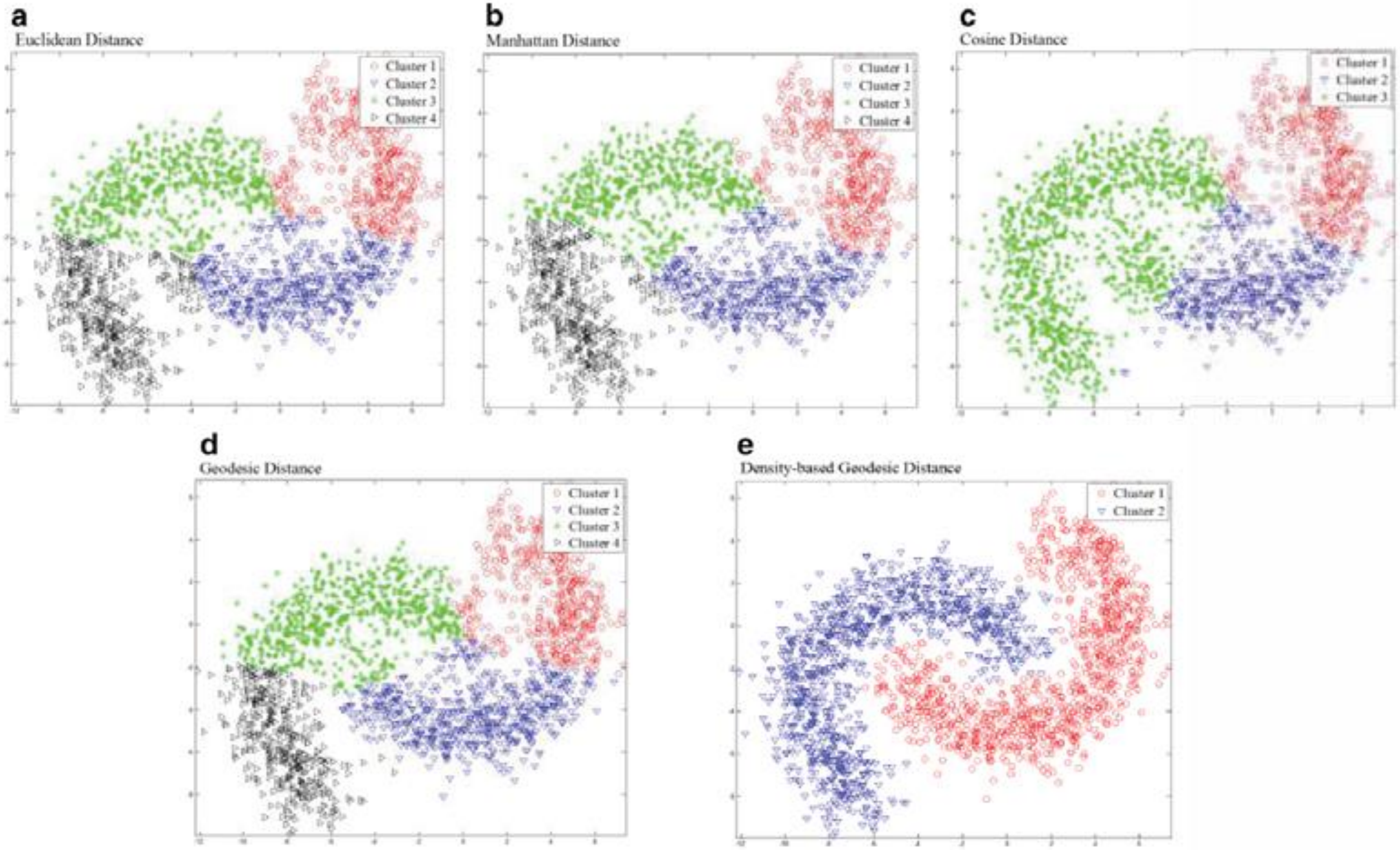
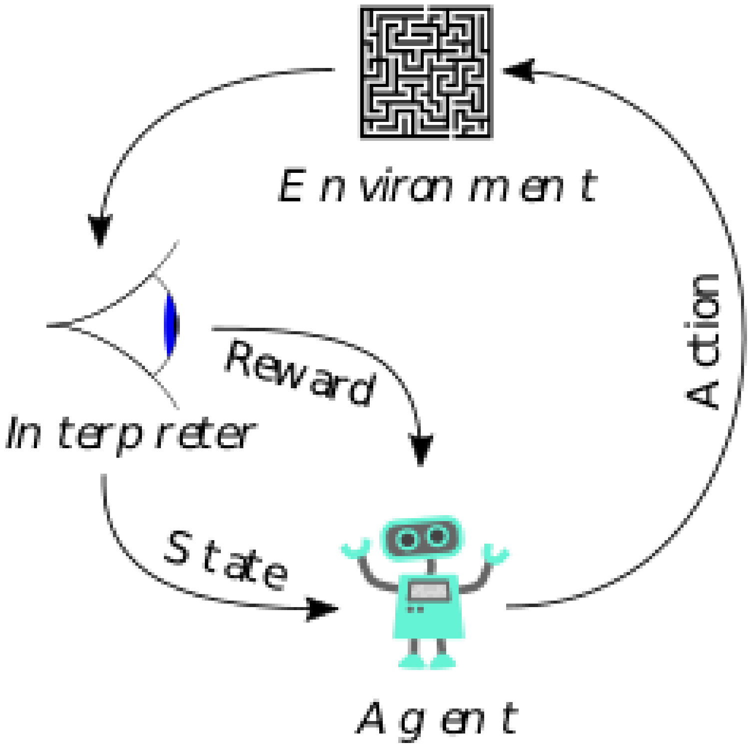


Fig. 7. PAM clustering results of Scenario 2: (a) Euclidean distance ($K = 4$), (b) Manhattan distance ($K = 4$), (c) cosine distance ($K = 3$), (d) geodesic distance ($k = 8$, $K = 4$), and (e) density-based geodesic distance ($k = 30$, $K = 2$).

Reinforcement Learning



- *Learning a policy: A sequence of outputs*
- *No supervised output but delayed reward*
- *Credit assignment problem*
- *Game playing*
- *Robot in a maze*
- *Multiple agents, partial observability, ...*

(sources: Ethem Alpaydın)

References

Introduction to Machine Learning with Python, a Guide for Data Scientists by Andreas C. Müller & Sarah Guido